

Migraine

Overview

Migraine is a common neurological disorder characterized by recurrent episodes of moderate to severe headache, often accompanied by nausea, vomiting, and increased sensitivity to light, sound, or smells. Unlike a typical headache, migraine is a complex neurological condition that can significantly interfere with daily activities, work, school, and quality of life.

Migraine affects people of all ages but is most commonly diagnosed in adolescents and adults. Women are affected more frequently than men, partly due to hormonal influences. Migraine attacks may last from a few hours to several days, and the frequency varies from person to person. Some individuals experience only a few attacks each year, while others may have several migraines each month.

Although migraine is not life-threatening, proper diagnosis and treatment are important to reduce symptoms and improve quality of life.

Types of Migraine

There are several types of migraine.

1. Migraine Without Aura

Migraine without aura is the most common form. Patients experience headache and associated symptoms without any warning signs before the attack.

Common characteristics include:

- Moderate to severe headache
- Usually affects one side of the head
- Throbbing or pulsating pain
- Pain worsens with physical activity

2. Migraine With Aura

Migraine with aura includes temporary neurological symptoms that occur before the headache begins.

Aura symptoms may include:

- Flashing lights
- Zigzag lines
- Blind spots
- Tingling sensation
- Difficulty speaking
- Temporary weakness

Aura usually lasts between 5 and 60 minutes before the headache starts.

3. Chronic Migraine

Chronic migraine is diagnosed when headache occurs on 15 or more days each month for at least three months, with migraine features present on at least eight of those days.

Patients with chronic migraine often experience significant disability and may require preventive treatment.

4. Vestibular Migraine

Vestibular migraine is characterized by episodes of dizziness or vertigo associated with migraine.

Symptoms include:

- Spinning sensation
- Loss of balance
- Motion sensitivity
- Nausea

Headache may or may not be present during these episodes.

Causes

The exact cause of migraine is not completely understood. It is believed to involve abnormal activity in the brain affecting nerves, blood vessels, and chemical messengers such as serotonin.

Migraine often runs in families, suggesting a genetic component.

Several factors can trigger migraine attacks in susceptible individuals.

Common Triggers

Migraine triggers vary among individuals.

Common triggers include:

- Emotional stress
- Anxiety
- Sleep deprivation
- Excessive sleep
- Bright lights
- Loud sounds
- Strong odors
- Hormonal changes
- Menstruation
- Dehydration
- Skipping meals
- Excessive caffeine
- Alcohol
- Chocolate
- Aged cheese
- Processed foods containing nitrates
- Weather changes
- Screen exposure

- Fatigue

Identifying personal triggers can help reduce migraine frequency.

Symptoms

Migraine symptoms differ between individuals but commonly include:

- Moderate to severe headache
- Throbbing pain
- One-sided headache
- Head pain that worsens with movement
- Nausea
- Vomiting
- Sensitivity to light (photophobia)
- Sensitivity to sound (phonophobia)
- Sensitivity to smells
- Blurred vision
- Dizziness
- Difficulty concentrating
- Neck stiffness
- Fatigue

Some patients may experience symptoms several hours before the headache begins, such as mood changes, food cravings, frequent yawning, or difficulty concentrating.

Migraine Phases

Migraine attacks may occur in four phases.

Prodrome

Symptoms before the headache may include:

- Mood changes
 - Fatigue
 - Food cravings
 - Frequent yawning
 - Neck stiffness
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Aura

Some patients experience temporary neurological symptoms including:

- Flashing lights
 - Blurred vision
 - Tingling sensation
 - Speech difficulty
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Headache Phase

Typical symptoms include:

- Severe throbbing headache
- Nausea
- Vomiting
- Light sensitivity
- Sound sensitivity

The headache usually lasts between 4 and 72 hours if untreated.

Postdrome

After the headache resolves, patients may experience:

- Tiredness
- Difficulty concentrating

- Muscle soreness
- General weakness

Recovery may take up to 24 hours.

Diagnosis

There is no single laboratory test for migraine.

Diagnosis is based on:

- Medical history
- Description of headache pattern
- Neurological examination
- Family history
- Identification of triggers

Brain imaging such as CT scan or MRI may be recommended if symptoms are unusual or if another neurological condition is suspected.

Treatment

Treatment aims to relieve symptoms during an attack and prevent future migraines.

Acute Treatment

Common medications include:

- Paracetamol
- Non-steroidal anti-inflammatory drugs (NSAIDs)
- Triptans
- Anti-nausea medications

Early treatment after symptom onset is generally more effective.

Preventive Treatment

Patients with frequent migraines may benefit from preventive therapy.

Options include:

- Beta blockers
- Calcium channel blockers
- Anti-epileptic medications
- Certain antidepressants
- CGRP inhibitors
- Botulinum toxin injections for chronic migraine

Treatment should be individualized based on the patient's symptoms and medical history.

Lifestyle Management

Lifestyle changes play an important role in migraine prevention.

Patients are encouraged to:

- Maintain regular sleep schedules
- Stay hydrated
- Eat balanced meals
- Avoid skipping meals
- Exercise regularly
- Manage stress through relaxation techniques
- Limit caffeine intake
- Keep a headache diary
- Identify and avoid personal triggers

These measures may reduce both the frequency and severity of migraine attacks.

Possible Complications

Although migraine itself is usually not life-threatening, complications may occur.

Possible complications include:

- Chronic migraine
- Medication-overuse headache
- Persistent aura
- Reduced quality of life
- Anxiety
- Depression
- Sleep disturbances

Patients with sudden changes in headache pattern or neurological symptoms should seek medical evaluation.

When to Seek Emergency Care

Medical attention should be sought immediately if headache is associated with:

- Sudden severe headache ("worst headache of life")
- Loss of consciousness
- Weakness of an arm or leg
- Facial drooping
- Slurred speech
- Persistent vomiting
- High fever
- Seizure
- Confusion
- Vision loss

These symptoms may indicate a more serious neurological condition such as stroke, meningitis, or brain hemorrhage.

Patient Advice

Patients should keep a record of headache frequency, duration, associated symptoms, and possible triggers. Regular follow-up with a healthcare provider can help optimize treatment and reduce migraine-related disability.

Medication should be taken only as prescribed to avoid medication-overuse headaches. Lifestyle modifications, trigger avoidance, and adherence to preventive therapy can significantly improve long-term outcomes.

Summary

Migraine is a chronic neurological disorder characterized by recurrent episodes of moderate to severe headache accompanied by symptoms such as nausea, vomiting, dizziness, and sensitivity to light or sound. While there is no permanent cure, appropriate medical treatment, preventive therapy, and healthy lifestyle habits can greatly reduce the frequency and severity of migraine attacks. Early recognition of symptoms and identification of individual triggers are essential for effective management.